

Nightmares Course: Seizure

Section 1: Case Summary

Scenario Title:	Nightmares Course: Seizure
Keywords:	Seizure, altered level of consciousness
Brief Description of Case:	The resident is called to the ward to manage a patient who may have had a seizure. The patient is somnolent when the resident arrives. Shortly afterward, the patient seizes again. Two doses of anti-epileptic will be required to terminate the seizure. Finally, when the patient has been stabilized, the resident will be required to discuss the case with their staff on call.

Goals and Objectives	
Educational Goal:	To review the approach to a seizing patient.
Objectives: (Medical and CRM)	1. Demonstrate an organized approach to the initial evaluation of a seizing patient. 2. Call for help in a timely manner. 3. Begin pharmacologic management of the seizure with benzodiazepines. 4. Consider possible etiologies of the seizure. 5. Communicate the clinical situation clearly to consultants and initiate a transfer to a higher level of care.
EPAs Assessed:	TTD1: Recognizing the unstable/critically ill patient, mobilizing the healthcare team, and initiating basic life support TTD3: Facilitating communication of information between a patient, caregivers, and members of the healthcare team to organize care and disposition of the patient F1: Initiating and assisting in resuscitation of critically ill patients

Learners, Setting and Personnel			
Target Learners:	☐ Medical Students	☒ Junior Residents	☐ Senior Residents
	☐ Staff Physicians	☐ Inter-professional	☐ Other
Location:	☒ Sim Lab	☐ In Situ	☐ Other
Recommended Number of Facilitators:	Instructors: 1-2		
	Confederates: 1		
	Sim Techs: 1		

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
Affiliations/Institutions(s):	Queen's University
Contact E-mail:	
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Revised By:	Dr. Kyla Caners
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: John Wright		Age: 67		Gender: Male	
Weight: 75kg					
Presenting complaint: Seizure					
Temp: 36.5	HR: 114	BP: 144/90	RR: 14	O ₂ Sat: 99%	FiO ₂ : r/a
Triage note: It is 01:00 and you are on call covering the thoracic surgery service. You have been called to assess Mr. Wright for a seizure episode.					
Allergies: None					
Past Medical History: Admitted awaiting lobectomy for non-small cell lung cancer.			Current Medications: Hydromorphone-contin 6mg BID Nicotine patch 21mg Multivitamins Gabapentin 300mg TID		

Section 2B: Extra Patient Information

A. Further History
To be provided by nurse (if asked): - Patient is full code - His primary issues in hospital have been pain control and his meds have been increased almost daily - He has no prior history of seizures Patient is unable to provide any history as he is too somnolent.
B. Physical Exam
<i>(List any pertinent findings – may be helpful to use ABCDE format.)</i> A – Airway is patent but there is blood in the mouth. B – Sonorous respirations. No increased work of breathing or other adventitious sounds C – Normal cardiac exam. D - Patient is somnolent and not answering any questions. He will moan to painful stimuli. There are no signs of trauma on exam. E – No signs of trauma and no lateralizing signs.

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
None.
C. Required Medications
Lorazepam and/or midazolam to be given IV.
D. Moulage
Blood in oropharynx. Hospital gown on, IV access in place.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
The patient will moan in response to pain. Otherwise, he does not respond to stimuli. The patient should have sonorous respirations. At about 1-2 minutes, the patient will begin seizing and will continue to seize until two anti-epileptics are given.

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside nurse	You are at the bedside to provide the learner with information about the patient and to assist as needed with tasks. You know that Mr. Wright has lung cancer and is awaiting a lobectomy. He is a full code. His primary issues in hospital have been pain control and (if asked), his meds have been increased almost daily. Depending on the mannequin being used, you may also have to cue the learners as to when the patient begins seizing and will need to continuously prompt about the seizures until the second benzodiazepine dose is given.
Senior resident	You will arrive 4-5 minutes into the case as ask the resident “what’s going on?” You will assist with arranging for transfer to a more acute ward, as needed. You can also provide guidance on anti-epileptic medications.

Simulation Scenario Template

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State GCS: E1V2M1 = 4. No vitals given in this state (patient seizes before they can be obtained).	Sonorous respirations, moans to painful stimuli.	<u>Expected Learner Actions</u> ☐ Brief history and physical ☐ Obtain information from bedside nurse ☐ Attempt vital signs	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - Vitals attempted → 2. Seizure	
2. Seizure Rhythm: sinus tach HR: 114 BP: 144/90 RR: 14 O ₂ SAT: 99% T: 36.5°C	Patient having generalized tonic/clonic seizure.	<u>Expected Learner Actions</u> ☐ Obtain blood glucose (6.2mmol/L) ☐ Lorazepam 2-4mg iv or midazolam 5mg iv ☐ Call for help ☐ Second dose of lorazepam or midazolam	<u>Modifiers</u> - 1 st dose benzodiazepine → no effect - Senior resident to arrive 4-5 min into case (after 1 st dose benzo given) <u>Triggers</u> - 2 nd dose benzo given → 3. Somnolence	
3. Somnolence Rhythm: sinus tach HR: 98 BP: 144/90 RR: 14 O ₂ SAT: 99% T: 36.5°C	Patient no longer seizing, but again sonorous. No focal neuro findings	<u>Expected Learner Actions</u> ☐ Obtain labs ☐ Consider CT head ☐ Discuss with senior about higher level care area ☐ Update attending	<u>Modifiers</u> - Thoracics attending to call 1-2 min into state for patient update <u>Triggers</u> - End case PRN after discussion with attending	



Simulation Scenario Template

Appendix A: Laboratory Results – None required during case.

<u>CBC</u> WBC Hgb Plt <u>Lytes</u> Na K Cl HCO ₃ AG Urea Cr Glucose <u>Extended Lytes</u> Ca Mg PO ₄ Albumin TSH <u>VBG</u> pH pCO ₂ pO ₂ HCO ₃ Lactate	<u>Cardiac/Coags</u> Trop D-dimer INR aPTT <u>Biliary</u> AST ALT GGT ALP Bili Lipase <u>Tox</u> EtOH ASA Tylenol Dig level Osmols <u>Other</u> B-HCG
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Simulation Scenario Template

Appendix B: ECGs, X-rays, Ultrasounds and Pictures – None Required for this case

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issue expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

The goal of this debriefing is to review the following:

1. The approach to the seizing patient.
2. Options for help during the night (senior resident, on call staff, ED attending, in house rapid response team).

Consider reviewing the ACEP guidelines for seizure management in the ED

<https://www.acep.org/globalassets/new-pdfs/clinical-policies/seizures-2014-final.pdf>

References

1. <https://www.acep.org/globalassets/new-pdfs/clinical-policies/seizures-2014-final.pdf>
- 2.
- 3.

